
Numerical Methods for Differential Equations



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Project Report
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ROOT FINDING ALGORITHMS

1

1.1 The concept or errors

1.2 Solution of linear equation

We solve equations of the form $f(x_1, \dots, x_n) = 0$ where this forms a linear system:

$$\begin{aligned} a_{11}x_1 + \dots + a_{1n}x_n - b_1 &= 0 \\ \vdots \\ a_{m1}x_1 + \dots + a_{mn}x_n - b_m &= 0 \end{aligned}$$

Further, we write $A = (a^1, \dots, a^n)_{m \times n}$. Recall, $A\vec{x} = \vec{b}$ is how we generally write it. Let us review the basic properties of solution based on dimensions of the matrices.

- $\rho(A) > \rho(A|b)$ is impossible!
- $\rho(A) < \rho(A|b) \implies$ no solution since $b \notin \text{Range}(a^1, \dots, a^n)$.
- $\rho(A) = \rho(A|b) = n \implies$ unique solution.
- $\rho(A) = \rho(A|b) < n \implies$ infinitely many solutions since $b \in \text{Range}(a^1, \dots, a^n)$ and some columns collapse.

Also since $\rho(A) \leq \min(m, n) \implies m < n$ and $n < m$ cases are covered already:

- $m < n$ (less equations, more unknowns): $\rho(A) \leq m < n$, so it no longer can have a unique solution. $\rho(A|b) > \rho(A) \implies$ no solution, and $\rho(A|b) = \rho(A) < n \implies$ infinitely many solutions.
- $m > n$ (more equations, less unknowns): $\rho(A) \leq n < m$. More equations could always be made by repeating equations in a system that has no solution / infinitely many / unique; so naturally for $m > n$ we have all possibilities.

We put $\rho(A|b)$ in various places relative to $\rho(A)$, n , and beyond n :

- $\rho(A) = \rho(A|b) < n \implies$ infinitely many solutions.
- $\rho(A) < \rho(A|b) \leq n \implies$ no solution.
- $\rho(A) < n = \rho(A|b) \implies$ no solution!
- $\rho(A) < n < \rho(A|b) \implies$ no solution.
- $n = \rho(A) = \rho(A|b) \implies$ unique solution.

1.2.1 Gauss-Elimination

We solve for a solution, when we have one, by Gauss elimination. This is the standard method we see in Linear Algebra. Let us see the number of floating-point operations (#FLOPs) a computer does in general for implementing this method.

$$\left[\begin{array}{cccc|c} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{array} \right] \xrightarrow[\alpha_1 = \frac{a_{21}}{a_{11}}]{R'_2 = R_2 - \alpha_1 R_1} \left[\begin{array}{cccc|c} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ 0 & a_{22}^{(1)} & \cdots & a_{2n}^{(1)} & b_2^{(1)} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & a_{m2}^{(1)} & \cdots & a_{mn}^{(1)} & b_m^{(1)} \end{array} \right]$$

One Gauss step requires:

- 1 extra multiplication ($\sim$ division) to get $\alpha_1 = \frac{a_{21}}{a_{11}}$.
- 1 A & 1 M (i.e., 1 addition/subtraction and 1 multiplication/division) to get $a_{2j}^{(1)}$ from a_{2j} for each $j = 2, \dots, n$ and b_2 , that is, n operations. The first 0 is not calculated at all and the machine simply purts it since making it 0 was the goal of $R_2 - \alpha_1 R_1$ operation to begin with.
- To get the entire first column zeroed, we do this $m - 1$ times. Hence, to get here we do $(m - 1)n$ additions and $(m - 1)(n + 1)$ multiplications.

Remark 1.1. Now, $a_{22}^{(1)}$ is our pivot entry—that is, we use it to make the column below it 0. Note that the matrix we wrote above is all that is there in computer memory, and all memory of original A is gone since keeping that in memory is pointless. In the following matrix, even though we write $a_{22}^{(1)}$ to separate it from a_{22} for us humans, there honestly isn't any need for that to a machine; $a_{22}^{(1)}$ is not only our new (2, 2)-th entry but also the only entry to the machine.

Next step:

$$\left[\begin{array}{ccccc|c} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} & b_1 \\ 0 & a_{22}^{(1)} & a_{23}^{(1)} & \cdots & a_{2n}^{(1)} & b_2^{(1)} \\ 0 & 0 & a_{33}^{(2)} & \cdots & a_{3n}^{(2)} & b_3^{(2)} \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & a_{m3}^{(2)} & \cdots & a_{mn}^{(2)} & b_m^{(2)} \end{array} \right]$$

Here, 1 A & 1 M are done $(n - 1)$ times ($a_{32}^{(1)} \rightarrow 0$ is known). Like before, for one row the number of FLOPs is $(n - 1)$ A and $(n - 1) + 1$ M, where +1 is due to the multiplication/division done to get the α factor for this step. Since there are $m - 2$ rows in total, the number of FLOPs is $(m - 2)(n - 1)$ additions and $(m - 2)n$ multiplications.

$$\left[\begin{array}{cccc|c} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ 0 & a_{22} & \cdots & a_{2n} & b_2 \\ 0 & 0 & \cdots & a_{3n} & b_3 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & a_{nn} & b_n \end{array} \right]$$

When reaching our target upper triangular matrix, note that if we are dealing with a rectangular matrix $n < m$ (or $m < n$), the process will stop before m (or n) steps because both $n - j$ and $m - i$ are involved and one of them will become 0. Naturally, for a rectangular matrix the number of calculations will be lower than for a square matrix. Let us assume $m = n$ to find the maximum #FLOPs.

#FLOPs for addition/subtraction (A):

$$\begin{aligned} \text{Total A} &:= (n - 1)n + (n - 2)(n - 1) + \cdots + 1 \cdot 2 \\ &= \sum_{j=1}^{n-1} j(j + 1) \sim \mathcal{O}\left(\frac{n^3}{3}\right) \end{aligned}$$

#FLOPs for multiplication/division (M):

$$\begin{aligned} \text{Total M} &:= (n - 1)(n + 1) + (n - 2)n + \cdots + 2 \cdot 4 + 1 \cdot 3 \\ &= \sum_{i=1}^{n-1} i(i + 2) \sim \mathcal{O}\left(\frac{n^3}{3}\right) \end{aligned}$$

If the number of variables increase by one, the order of effort the machine puts for finding the solution increases by $\frac{2}{3}n^3$. That is a lot of machine effort being used. Moreover, we are not even done yet! We further do the backward substitution to get the solution $\vec{x}$:

$$\begin{aligned} x_n &= \frac{b_n}{a_{nn}} \longrightarrow 0 \text{ A, } 1 \text{ M} \\ a_{n-1,n-1}x_{n-1} + a_{n-1,n}x_n &= b_{n-1} \implies x_{n-1} = \frac{1}{a_{n-1,n-1}} \left(b_{n-1} - a_{n-1,n} \frac{b_n}{a_{nn}} \right) \longrightarrow 1 \text{ A, } 2 \text{ M} \\ \text{Similarly, } x_{n-2} &= \frac{1}{a_{n-2,n-2}} (b_{n-2} - a_{n-2,n-1}x_{n-1} - a_{n-2,n}x_n) \longrightarrow 2 \text{ A, } 3 \text{ M} \\ &\vdots \end{aligned}$$

$$\text{FLOPs} := \sum_{j=0}^{n-1} [j + (j+1)] = \sum_{j=0}^{n-1} (2j+1) \sim \mathcal{O}\left(\frac{n^2}{2}\right)$$

The overall effort to solve $A\vec{x} = \vec{b}$ by Gauss-Elimination method is $\frac{n^2}{2} + \frac{2}{3}n^3$.

We understand already that we need to come up with better algorithms for practical uses. Before that, we will see a modification of Gauss-Elimination that works on special matrices called **Banded matrices**. They show up all the time in engineering (such as the heat equation, Black-Scholes, etc.), or more generally, whenever we discretize a differential equations take place and end up with a dependency on the previous and next steps. This is the **Thomas Algorithm**.

$$\left[\begin{array}{cccccc|c} a_{11} & a_{12} & 0 & 0 & \cdots & 0 & b_1 \\ a_{21} & a_{22} & a_{23} & 0 & \cdots & 0 & b_2 \\ 0 & a_{32} & a_{33} & a_{34} & \cdots & 0 & b_3 \\ 0 & 0 & a_{43} & a_{44} & \cdots & 0 & b_4 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & a_{nn} & b_n \end{array} \right] \longrightarrow \left[\begin{array}{cccccc|c} 1 & a_{12}^{(1)} & 0 & 0 & \cdots & 0 & b_1^{(1)} \\ 0 & a_{22}^{(1)} & a_{23} & 0 & \cdots & 0 & b_2^{(1)} \\ 0 & a_{32} & a_{33} & a_{34} & \cdots & 0 & b_3 \\ 0 & 0 & a_{43} & a_{44} & \cdots & 0 & b_4 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & a_{nn} & b_n \end{array} \right]$$

- We divide the first row by a_{11} . So, we get $\{1, a_{12}/a_{11} \sim a_{12}^{(1)}, 0, \dots, b_1/a_{11} \sim b_1^{(1)}\}$; hence we only perform 2 M FLOPs (because the first 1 is simply placed there).
- In step 2, we use 1 to make $a_{21} \rightarrow 0$. We simply put 0 there. To get $a_{22}^{(1)}$ from a_{22} and $b_2^{(1)}$ from b_2 , we do 1 M and 1 A each. There are no changes in a_{23} since the entry above it is 0. Total changes: 2 M and 2 A.
- In step 3, the third row will also have 1 M and 1 A each, but something changes from this row onward:

$$a_{33}^{(1)} = a_{33} - c a_{23}, \quad \text{where } c = \frac{a_{32}}{a_{22}^{(1)}}$$

This division was not needed earlier since we had 1 in a_{11} . Row 3 onwards will thus require an extra M.

$$\begin{array}{ll} \text{1st row:} & 0 \text{ A} \quad 2 \text{ M} \\ \text{2nd row:} & 2 \text{ A} \quad 2 \text{ M} \\ \text{3rd row:} & 2 \text{ A} \quad 2 \text{ M} + 1 \text{ M} \\ & \vdots \\ \text{n-th row:} & 2 \text{ A} \quad 2 \text{ M} + 1 \text{ M} \end{array}$$

$$\text{Total additions/subtractions} = 2(n-1) = 2n-2$$

$$\text{Total multiplications/divisions} = 3(n-2) + 4 = 3n-2$$

$$\text{Total \#FLOPs} = (2n-2) + (3n-2) = 5n-4 \sim \mathcal{O}(5n) \quad (\text{linear only!})$$

This concludes the discussion on what we call the **Direct method**. From linear algebra, this is THE method learned to solve $A\vec{x} = \vec{b}$.

1.2.2 Gauss–Seidel and Jacobi methods

Now, we consider another class of methods: **Indirect methods / Iterative methods**. These are the improved algorithm for machine we were talking about earlier.

- Gauss–Seidel Method
- Jacobi Method

Both methods are very similar and geometrically motivated. We start with a simple example of a system with 2 variables and 2 unknowns:

$$\begin{cases} x_1 + 2x_2 = 1 \\ x_1 - x_2 = 4 \end{cases} \quad (1.1)$$

Rearrangement 1: Let us start with first equation and make x_1 the subject of the 1st equation, then x_2 the subject of the 2nd:

$$\begin{aligned} x_1 &= 1 - 2x_2 \\ x_2 &= x_1 - 4 \end{aligned}$$

Rearrangement 2: Let us again start with the second equation now and make x_1 the subject of this equation, then x_2 will be the subject of the other:

$$\begin{aligned} x_1 &= 4 + x_2 \\ x_2 &= \frac{1 - x_1}{2} \end{aligned}$$

Firstly, note that there are two more rearrangements. We pick one of the equations and pick which variable will be our subject; this choice matters to us.

Let the initial guess be $(x_1^{(0)}, x_2^{(0)})$.

Let us start with **Rearrangement 1** written as an iterative algorithm:

$$\begin{cases} x_1^{(i+1)} = 1 - 2x_2^{(i)} \\ x_2^{(i+1)} = x_1^{(i+1)} - 4 \end{cases}$$

Put $x_2^{(0)}$ to get $x_1^{(1)}$. Then put $x_1^{(1)}$ to get $x_2^{(1)}$ in the 2nd equation (hence the algorithm has a lag in the 1st equation, but none in the 2nd). We will now apply this iteratively. Put $x_2^{(1)}$ in the 1st equation again to get $x_1^{(2)}$, then put $x_1^{(2)}$ into the 2nd to get $x_2^{(2)}$, and so on:

$$(x_1^{(0)}, x_2^{(0)}) \longrightarrow (x_1^{(1)}, x_2^{(0)}) \longrightarrow (x_1^{(1)}, x_2^{(1)}) \longrightarrow (x_1^{(2)}, x_2^{(1)}) \longrightarrow (x_1^{(2)}, x_2^{(2)}) \longrightarrow \dots$$

At the end of every full cycle, we get an updated candidate for the solution:

$$\begin{aligned} (0, 0) &\implies x_1 = 1 - 0 = 1 \implies x_2 = 1 - 4 = -3 \longrightarrow (1, -3) \\ (1, -3) &\implies x_1 = 1 + 6 = 7 \implies x_2 = 7 - 4 = 3 \longrightarrow (7, 3) \\ (7, 3) &\implies x_1 = 1 - 6 = -5 \implies x_2 = -5 - 4 = -9 \longrightarrow (-5, -9) \\ (-5, -9) &\implies x_1 = 1 + 18 = 19 \implies x_2 = 19 - 4 = 15 \longrightarrow \text{not converging!} \end{aligned}$$

Something went wrong! For now, let's try the same algorithm with **Rearrangement 2** and see if it works out:

$$\begin{cases} x_1^{(i+1)} = 4 + x_2^{(i)} \\ x_2^{(i+1)} = \frac{1 - x_1^{(i+1)}}{2} \end{cases}$$

$$(0, 0) \implies x_1 = 4 \implies x_2 = \frac{1 - 4}{2} = -1.5 \longrightarrow (4, -1.5)$$

$$(4, -1.5) \implies x_1 = 4 - 1.5 = 2.5 \implies x_2 = \frac{1 - 2.5}{2} = -0.75 \longrightarrow (2.5, -0.75)$$

$$(2.5, -0.75) \implies x_1 = 4 - 0.75 = 3.25 \implies x_2 = \frac{1 - 3.25}{2} = -1.125 \longrightarrow (3.25, -1.125)$$

This sequence actually converges to the true solution $(3, -1)$.

Keep in mind that in reality, the true solution is not known beforehand; so if the process is converging or diverging needs to be addressed properly; Also, it is unknown which rearrangement will work for us and that is determined by something called the **Diagonal Dominance condition**. We will come to that soon. The stopping condition is determined by a preset tolerance error margin ε_a (approximate relative error).

Figure 1.1: Divergence of Rearrangement 1 in Gauss–Seidel Method

Figure 1.2: Convergence of Rearrangement 2 in Gauss–Seidel Method

Geometrically, Gauss–Seidel operates as follows: Starting from the initial guess $(x_1^{(0)}, x_2^{(0)})$, Rearrangement 1 hits the first line (the equation with lag) and then alternates between the lines, diverging outward. Rearrangement 2 hits the line corresponding to its first update and spirals inwards to the intersection point $(3, -1)$. There are in total four traversal possibilities depending on which line and which axis direction is updated first:

- Hit the blue line along the x_1 axis.
- Hit the blue line along the x_2 axis.
- Hit the red line along the x_1 axis.
- Hit the red line along the x_2 axis.

To standardize the algorithm, we fix the update order: $x_1 \rightarrow x_2$.

Before stating the general n -variable scheme, consider the 3-variable case:

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 = b_1 \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 = b_2 \\ a_{31}x_1 + a_{32}x_2 + a_{33}x_3 = b_3 \end{cases}$$

Rewriting:

$$\begin{aligned}x_1 &= \frac{1}{a_{11}}(b_1 - a_{12}x_2 - a_{13}x_3) \\x_2 &= \frac{1}{a_{22}}(b_2 - a_{21}x_1 - a_{23}x_3) \\x_3 &= \frac{1}{a_{33}}(b_3 - a_{31}x_1 - a_{32}x_2)\end{aligned}$$

With initial guess $(x_1^{(0)}, x_2^{(0)}, x_3^{(0)})$ and update sequence $x_1 \rightarrow x_2 \rightarrow x_3$:

- Put $x_2^{(0)}, x_3^{(0)}$ to get $x_1^{(1)}$.
- Use new $x_1^{(1)}$ and old $x_3^{(0)}$ to get $x_2^{(1)}$.
- Use new $x_1^{(1)}$ and new $x_2^{(1)}$ to get $x_3^{(1)}$.

Then treat $(x_1^{(1)}, x_2^{(1)}, x_3^{(1)})$ as the new initial point and repeat:

$$\begin{aligned}(0, 0, 0) &\rightarrow (x_1^{(1)}, 0, 0) \rightarrow (x_1^{(1)}, x_2^{(1)}, 0) \rightarrow (x_1^{(1)}, x_2^{(1)}, x_3^{(1)}) \\(x_1^{(1)}, x_2^{(1)}, x_3^{(1)}) &\rightarrow (x_1^{(2)}, x_2^{(1)}, x_3^{(1)}) \rightarrow (x_1^{(2)}, x_2^{(2)}, x_3^{(1)}) \rightarrow (x_1^{(2)}, x_2^{(2)}, x_3^{(2)}) \rightarrow \dots\end{aligned}$$

The General n -variable Gauss–Seidel Scheme: For $\sum_{j=1}^n a_{ij}x_j = b_i, i = 1, \dots, n$:

$$\boxed{x_i^{(m+1)} = \frac{1}{a_{ii}} \left(b_i - \sum_{j=1}^{i-1} a_{ij}x_j^{(m+1)} - \sum_{j=i+1}^n a_{ij}x_j^{(m)} \right), \quad i = 1, \dots, n} \quad (1.2)$$

This is well-defined since $x_1^{(m+1)}, \dots, x_{i-1}^{(m+1)}$ are already computed when calculating $x_i^{(m+1)}$. We continue until the step change satisfies a prescribed tolerance.

The General n -variable Jacobi Scheme: The Jacobi scheme updates all components simultaneously using only values from the previous iteration:

$$\boxed{x_i^{(m+1)} = \frac{1}{a_{ii}} \left(b_i - \sum_{j=1}^{i-1} a_{ij}x_j^{(m)} - \sum_{j=i+1}^n a_{ij}x_j^{(m)} \right), \quad i = 1, \dots, n} \quad (1.3)$$

For $n = 3$:

$$\begin{aligned}x_1^{(1)} &= \frac{1}{a_{11}}(b_1 - a_{12}x_2^{(0)} - a_{13}x_3^{(0)}) \\x_2^{(1)} &= \frac{1}{a_{22}}(b_2 - a_{21}x_1^{(0)} - a_{23}x_3^{(0)}) \quad (\text{does not use updated } x_1) \\x_3^{(1)} &= \frac{1}{a_{33}}(b_3 - a_{31}x_1^{(0)} - a_{32}x_2^{(0)}) \quad (\text{does not use updated } x_1, x_2)\end{aligned}$$

While Gauss–Seidel uses updated coordinates immediately and moves in zig-zag pattern along the coordinate directions. Jacobi computes all entries from the old point before updating the full state vector and then takes a diagonal jump.

Figure 1.3: Gauss-Seidel vs Jacobi with three hyperplanes in $\mathbb{R}^3$

We will now see how we determine what specific “rearrangement” makes these processes converge. We already have the order of movement because of how the equations are defined in algorithms 1.2 and 1.3:

$$x_1 \longrightarrow x_2 \longrightarrow \cdots \longrightarrow x_n$$

Now the i^{th} equation $\sum_j a_{ij}x_j - b_i = 0$ defines the hyperplane H_i . We want to figure out the optimal order to hit these planes, e.g. , $\{H_2, H_4, H_3, \dots\}$ might be better than the obvious $\{H_1, H_2, H_3, \dots\}$

Definition 1.2. Strict Diagonal Dominance (SDD) : A matrix $A = (a_{ij})_{i,j} \in \mathbb{R}^{n \times n}$ is strictly diagonally dominant if the absolute value of the diagonal entry is strictly larger than the sum of the absolute values of all other entries in that row:

$$|a_{ii}| > \sum_{j \neq i} |a_{ij}| \quad \text{for all } i = 1, \dots, n \quad (1.4)$$

This algebraic condition guarantees:

1. $\det(A) \neq 0$
2. Gauss–Seidel and Jacobi methods converge.

In our earlier example 1.1, recall:

$$\begin{cases} x_1 + 2x_2 = 1 \\ x_1 - x_2 = 4 \end{cases} \longrightarrow \text{ordering did not work out (not diagonally dominant)}$$

$$\begin{cases} x_1 - x_2 = 4 \\ x_1 + 2x_2 = 1 \end{cases} \longrightarrow \text{ordering worked (diagonally dominant)}$$

We can reshuffle rows of the matrix A and try to get a diagonally dominant matrix.

Geometric Intuition : Recall the linear system $Ax = \mathbf{b}$, where the i^{th} equation

$$a_{ii}x_i + \sum_{j \neq i} a_{ij}x_j = b_i$$

defines a hyperplane H_i with normal vector $\mathbf{n}_i = (a_{i1}, a_{i2}, \dots, a_{ii}, \dots, a_{in})^T$.

In stationary iterative methods (such as Jacobi or Gauss–Seidel), the i^{th} step updates the coordinate x_i by moving parallel to the standard basis vector $\mathbf{e}_i$ until intersecting H_i .

A large diagonal magnitude $|a_{ii}| > \sum_{j \neq i} |a_{ij}|$ implies that the normal vector $\mathbf{n}_i$ is predominantly aligned with $\mathbf{e}_i$. Geometrically, this means H_i is nearly orthogonal to the direction of motion $\mathbf{e}_i$ (and nearly parallel to the coordinate hyperplane $\{x_i = 0\}$). Consequently, advancing along $\mathbf{e}_i$ strikes H_i transversely with minimal displacement.

Conversely, if $|a_{ii}|$ is small, $\mathbf{n}_i$ is almost orthogonal to $\mathbf{e}_i$, rendering H_i nearly parallel to the line of motion. Stepping along x_i to enforce the i^{th} constraint then produces an enormous displacement $|\Delta x_i|$, throwing the iterate far away from the true solution and triggering divergence.

Recall example 1.1, $H_1 : x_1 + 2x_2 = 1$ with normal $\mathbf{n}_1 = (1, 2)^T$, and $H_2 : x_1 - x_2 = 4$ with normal $\mathbf{n}_2 = (1, -1)^T$. Because $\mathbf{n}_1$ is dominantly aligned with the x_2 -axis, we know H_1 is almost parallel to x_1 . Moving along x_1 will take us too far!

Figure 1.4: Hyperplane orientation and its normal vector

Figure 1.5: Hyperplanes H_1 and H_2 and their normal vectors

Summary: If one of the hyperplanes we get from the equation is almost parallel to one of the standard hyperplanes $\{x_i = 0\}$, then Strict Diagonal Dominance ensures we handle it at i^{th} iteration when we are moving along $\vec{e}_i$.

Theorem 1.3. *Levy-Desplanques Theorem : If A is SDD, then $\det(A) \neq 0$.*

Proof. Firstly, we are going to need this in the convergence results of Gauss-Seidel and Jacobi. Secondly, we proceed by contradiction. Suppose that $\det(A) = 0$. Then the linear system $A\vec{x} = \vec{0}$ admits a non-trivial solution. That is, there exists a vector $\vec{x} = (x_1, x_2, \dots, x_n)^T \in \mathbb{R}^n$ such that $\vec{x} \neq \vec{0}$ and

$$A\vec{x} = \vec{0}.$$

Since $\vec{x} \neq \vec{0}$, the set of absolute values $\{|x_1|, |x_2|, \dots, |x_n|\}$ contains at least one strictly positive element. Let $k \in \{1, 2, \dots, n\}$ be an index corresponding to a component of maximal magnitude:

$$|x_k| = \max_{1 \leq j \leq n} |x_j| > 0.$$

Now consider the k -th equation of the homogeneous system $A\vec{x} = \vec{0}$:

$$\sum_{j=1}^n a_{kj}x_j = 0 \implies a_{kk}x_k = -\sum_{\substack{j=1 \\ j \neq k}}^n a_{kj}x_j \quad (\text{isolating the diagonal term})$$

$$\text{i.e., } |a_{kk}||x_k| = \left| -\sum_{\substack{j=1 \\ j \neq k}}^n a_{kj}x_j \right| \leq \sum_{\substack{j=1 \\ j \neq k}}^n |a_{kj}||x_j|$$

By our choice of k , we have $|x_j| \leq |x_k|$ for all $j \in \{1, \dots, n\}$. Therefore,

$$|a_{kk}||x_k| \leq \sum_{\substack{j=1 \\ j \neq k}}^n |a_{kj}||x_k| = |x_k| \sum_{\substack{j=1 \\ j \neq k}}^n |a_{kj}|.$$

Since $|x_k| > 0$, we may divide both sides by $|x_k|$ to arrive at:

$$|a_{kk}| \leq \sum_{\substack{j=1 \\ j \neq k}}^n |a_{kj}|.$$

This directly contradicts the SDD condition (1.2) for row k . $\square$

This consequence of SDD is pretty geometric and natural as compared to the other two. We have already established how SDD means i^{th} row is predominantly aligned with the i^{th} coordinate axis for each i . Naturally, the rows will span a n -volume element in $\mathbb{R}^n$. Hence, $\det(A) \neq 0$

Figure 1.6: n -volume element in $\mathbb{R}^n$ for SDD matrix

We can now focus on the main results about convergence of our schemes. An iterative method of solving comprises, firstly, of writing:

$$A = M - N$$

where M is invertible and $N = M - A$. Then,

$$\begin{aligned} A\vec{x} = \vec{b} &\implies (M - N)\vec{x} = \vec{b} \\ &\implies M\vec{x} = N\vec{x} + \vec{b} \\ &\implies \vec{x} = M^{-1}N\vec{x} + M^{-1}\vec{b} \end{aligned}$$

We then write this as an iterative algorithm:

$$\vec{x}^{(m+1)} = (M^{-1}N)\vec{x}^{(m)} + M^{-1}\vec{b} \quad (1.5)$$

and we call $M^{-1}N$ the iteration matrix for the process.

For Gauss–Seidel and Jacobi we simply have to pick an appropriate invertible M and then figure out N by $N = M - A$ to get the iteration matrices T_G and T_J respectively. Since there is component-wise updating involved in every step, T_G is a bit more involved. Let's see T_J first.

Theorem 1.4. *If A is SDD, then Jacobi methods converges.*

Proof. Write $A = D - L - U$, where D is the diagonal component, $-L$ is strictly lower triangular, and $-U$ is strictly upper triangular such that

$$\begin{aligned} A &= D - (L + U) \\ &\implies D^{-1}A + D^{-1}(L + U) = I \\ &\implies D^{-1}A\vec{x} + D^{-1}(L + U)\vec{x} = \vec{x} \\ &\implies D^{-1}\vec{b} + D^{-1}(L + U)\vec{x} = \vec{x} \end{aligned}$$

Let Jacobi iteration matrix be

$$T_J = D^{-1}(L + U) \quad (1.6)$$

D^{-1} exists since diagonal entries are non-zero since $a_{ii} \neq 0$ for any i . If not, that entire row is 0 by SDD and then, by Theorem 1.3, A cannot be SDD!

Iterative algorithm:

$$\vec{x}^{(m+1)} := D^{-1}\vec{b} + T_J\vec{x}^{(m)}, \quad \text{where } T_J = D^{-1}(L + U)$$

Let $\vec{x}^*$ be the true solution $\implies D^{-1}\vec{b} + T_J\vec{x}^* = \vec{x}^*$. Define error $\vec{e}^{(m)} = \vec{x}^{(m)} - \vec{x}^*$.

$$\begin{aligned} \vec{e}^{(m+1)} &= \vec{x}^{(m+1)} - \vec{x}^* \\ &= D^{-1}\vec{b} + T_J\vec{x}^{(m)} - \vec{x}^* \\ &= D^{-1}\vec{b} + T_J(\vec{e}^{(m)} + \vec{x}^*) - \vec{x}^* \\ &= D^{-1}\vec{b} + T_J\vec{e}^{(m)} + T_J\vec{x}^* - \vec{x}^* \\ &= T_J\vec{e}^{(m)} + (D^{-1}\vec{b} + T_J\vec{x}^* - \vec{x}^*) \\ &= T_J\vec{e}^{(m)} \end{aligned}$$

Then,

$$\vec{e}^{(m)} = T_J^m \vec{e}^{(0)} \tag{1.7}$$

So, if the spectral radius $\rho(T_J) < 1$, then $\|\vec{e}^{(m)}\| \rightarrow 0$ as $m \rightarrow \infty$.

Structure of T_J :

$$\begin{aligned} T_J &= \begin{bmatrix} 0 & -\frac{a_{12}}{a_{11}} & -\frac{a_{13}}{a_{11}} & \cdots & -\frac{a_{1n}}{a_{11}} \\ -\frac{a_{21}}{a_{22}} & 0 & -\frac{a_{23}}{a_{22}} & \cdots & -\frac{a_{2n}}{a_{22}} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ -\frac{a_{n1}}{a_{nn}} & -\frac{a_{n2}}{a_{nn}} & -\frac{a_{n3}}{a_{nn}} & \cdots & 0 \end{bmatrix} \\ \implies (T_J)_{ij} &= \begin{cases} -\frac{a_{ij}}{a_{ii}} & \text{for } i \neq j \\ 0 & \text{for } i = j \end{cases} \end{aligned}$$

Then taking the matrix infinity norm (maximum row sum):

$$\|T_J\|_\infty = \max_{1 \leq i \leq n} \sum_{j=1}^n |(T_J)_{ij}| = \max_{1 \leq i \leq n} \sum_{j \neq i} \frac{|a_{ij}|}{|a_{ii}|} < 1 \quad (\text{SDD})$$

Since $\rho(T_J) \leq \|T_J\|_\infty < 1$, convergence is guaranteed. $\square$

In Gauss-Seidel, as soon as we get $x_i^{(m+1)}$, we update the i^{th} coordinate to solve for the rest of the coordinates $x_{i+1}^{(m+1)}, \dots, x_n^{(m+1)}$. The decomposition $A = D - L - U$ is always true, but notice for Gauss-Seidel algorithm previous clubbing will not work.

$$a_{ii}x_i^{(m+1)} + \sum_{j < i} a_{ij}x_j^{(m+1)} = b_i - \sum_{j > i} a_{ij}x_j^{(m)}$$

Here $a_{ii}x_i^{(m+1)}$ corresponds to $D\vec{x}^{(m+1)}$, $\sum_{j < i} a_{ij}x_j^{(m+1)}$ corresponds to $-L\vec{x}^{(m+1)}$, and $\sum_{j > i} a_{ij}x_j^{(m)}$ corresponds to $U\vec{x}^{(m)}$.

So, our decomposition here is: $A = \underbrace{(D - L)}_M - \underbrace{U}_N$

The Gauss-Seidel iterative matrix is

$$T_G = (D - L)^{-1}U \quad (1.8)$$

Here $(D - L)^{-1}$ is defined because $D - L$ is still a lower triangular matrix and determinant is same as that of D .

Theorem 1.5. *If A is SDD, then Gauss-Seidel methods converges.*

Proof. Like the Jacobi case, we show the spectral radius is strictly less than 1. Let λ be any eigenvalue and $\vec{v}$ be the corresponding eigenvector for T_G . Let v_k be the coordinate such that $|v_k| = \max_i |v_i|$.

$$\begin{aligned} T_G \vec{v} &= \lambda \vec{v} \\ \implies (D - L)^{-1}U \vec{v} &= \lambda \vec{v} \\ \implies U \vec{v} &= \lambda(D - L) \vec{v} \\ \implies (\lambda D - \lambda L - U) \vec{v} &= \vec{0} \end{aligned}$$

Row-wise representation for the i^{th} component:

$$\lambda a_{ii} v_i + \lambda \sum_{j < i} a_{ij} v_j + \sum_{j > i} a_{ij} v_j = 0 \quad \text{for all } i = 1, \dots, n$$

Look at v_k in particular and take absolute values:

$$\begin{aligned} |\lambda| |a_{kk}| |v_k| &= \left| -\lambda \sum_{j < k} a_{kj} v_j - \sum_{j > k} a_{kj} v_j \right| \leq |\lambda| \sum_{j < k} |a_{kj}| |v_j| + \sum_{j > k} |a_{kj}| |v_j| \\ \implies |\lambda| |a_{kk}| &\leq |\lambda| \sum_{j < k} |a_{kj}| \frac{|v_j|}{|v_k|} + \sum_{j > k} |a_{kj}| \frac{|v_j|}{|v_k|} \\ &\leq |\lambda| \sum_{j < k} |a_{kj}| + \sum_{j > k} |a_{kj}| \quad \left(\text{since } \frac{|v_j|}{|v_k|} \leq 1 \right) \\ \implies |\lambda| \left(|a_{kk}| - \sum_{j < k} |a_{kj}| \right) &\leq \sum_{j > k} |a_{kj}| \end{aligned}$$

Since A is SDD, $|a_{kk}| - \sum_{j < k} |a_{kj}| > \sum_{j > k} |a_{kj}| \geq 0$. Since $\vec{v} \neq \vec{0} \implies \max_i |v_i| \neq 0 \implies v_k \neq 0$:

$$|\lambda| \leq \frac{\sum_{j > k} |a_{kj}|}{|a_{kk}| - \sum_{j < k} |a_{kj}|}$$

Using SDD again:

$$\begin{aligned} |a_{kk}| &> \sum_{j \neq k} |a_{kj}| = \sum_{j < k} |a_{kj}| + \sum_{j > k} |a_{kj}| \\ \implies |a_{kk}| - \sum_{j < k} |a_{kj}| &> \sum_{j > k} |a_{kj}| \\ \implies |\lambda| &\leq \frac{\sum_{j > k} |a_{kj}|}{|a_{kk}| - \sum_{j < k} |a_{kj}|} < 1 \end{aligned}$$

Therefore, $\rho(T_G) < 1$. □

1.3 Solution of non-linear equation

Here we consider $f(x_1, \dots, x_n) = (0, \dots, 0) \in \mathbb{R}^m$, where equations contain non-linear terms such as polynomials, $\ln(\cdot)$, $\sin(\cdot)$, $\cos(\cdot)$, etc. This introduces systems that can have multiple isolated roots:

$$f(\vec{x}) = \begin{bmatrix} f_1(\vec{x}) \\ \vdots \\ f_m(\vec{x}) \end{bmatrix} = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix}$$

where f_i is a non-linear function for each i . Hence, it is necessary to start with the single-variable case $f(x) = 0$.

1. **Bracketing Methods** (require two initial guesses that bracket a root):

- Bisection Method
- Regula-Falsi Method

2. **Open Methods:**

- Newton-Raphson Method
- Secant Method
- Fixed-Point Iteration

1.3.1 Bisection method

Assume f is continuous in $[a, b]$ and also $f(a)f(b) < 0$, then it will have one root in $[a, b]$. That is the entire idea and this comes from Intermediate Value Theorem. We start with two points x_1, x_2 and if we have a sign change, we have a root since it must cut the x -axis somewhere. Then pick the middle point $x_0 = \frac{x_1+x_2}{2}$ and look for a sign change in $[x_1, x_0]$ and $[x_0, x_2]$.

Bisection method is effective but convergence is slow.

Define error as the difference between upper & lower bound of the root (i.e., size of interval in which we are looking for sign change). Then at every iteration, the error is halved & hence convergence rate is linear.

Figure 1.7: Sign change and interval halving in Bisection method

At k^{th} step error is $|b - a|$, then in $(k + 1)^{\text{th}}$ step it is $\frac{|b-a|}{2}$, and in $(k + 2)^{\text{th}}$ step it is

$\frac{|b-a|}{4}$, i.e.,

$$1^{\text{st}} \text{ step error} = l$$

$$2^{\text{nd}} \text{ step error} = \frac{l}{2}$$

$$3^{\text{rd}} \text{ step error} = \frac{2^{\text{nd}} \text{ step error}}{2}$$

$\vdots$

$$k^{\text{th}} \text{ step error} = \frac{(k-1)^{\text{th}} \text{ step error}}{2} \Rightarrow \mathcal{E}_k = \frac{1}{2} \mathcal{E}_{k-1} \quad \text{which is linear.}$$

Example 1.6. Let f be a polynomial,

$$f(x) = a_n x^n + \cdots + a_1 x + a_0$$

The range of the root is given by $[-x_{\max}, x_{\max}]$ and we will initiate bisection with $-x_{\max}$ and $x_{\max}$. From relation between roots & coefficients we know,

$$\sum x_i = -\frac{a_{n-1}}{a_n}, \quad \sum_{i < j} x_i x_j = \frac{a_{n-2}}{a_n}$$

Since,

$$(x_1 + \cdots + x_n)^2 = \sum_{i \neq j} x_i x_j + \sum x_i^2$$

$$\text{i.e., } \left(\sum x_i \right)^2 - 2 \sum_{i < j} x_i x_j = \sum x_i^2 := x_{\max}^2 (\text{let})$$

$$\text{i.e., } x_{\max} = \sqrt{\sum x_i^2}$$

$$\text{i.e., } x_{\max} = \sqrt{\left(\frac{a_{n-1}}{a_n} \right)^2 - 2 \left(\frac{a_{n-2}}{a_n} \right)}$$

$x_{\max}$ is our bound for roots because clearly for any root x_k , $x_k \leq \sqrt{\sum x_i^2}$.

Example 1.7. Consider the polynomial $f(x) = x^2 - 4x - 10$. Then we have,

$$x_{\max} = \sqrt{\left(-\frac{4}{1} \right)^2 - 2 \left(-\frac{10}{1} \right)} = \sqrt{36} = 6.$$

x	-6	-5	-4	-3	-2	-1	0	1	2	3	4	5	6
$f(x)$	50	35	22	11	2	-5	-10	-13	-14	-13	-10	-5	2

We have at least one root in $[-2, -1]$ and in $[5, 6]$.

Let's apply our algorithm in $[-2, -1]$ only.

Midpoint of $[-2, -1]$ is -1.5 and $f(-1.5) = -1.75 < 0$.

So, sign change is happening in $[-2, -1.5]$ instead of in $[-1.5, -1]$. We can further zoom in on $[-2, -1.5]$ interval. The midpoint is -1.75 and $f(-1.75) = 0.0625 > 0$.

x	$f(x)$
-2	2
-1.75	0.0625
-1.50	-1.75
-1.25	need not calculate
-1	-5

Root lies in $[-1.75, -1.50]$. Using computer program now we can get as close to the root as we want since length of the interval is getting halved every time. After few iterations approximate root is -1.7416 .

Example 1.8. Solving $2 - x + \ln(x) = 0$ (Excel codes)

A #	B $x(l)$	C $x(r)$	D $f(l)$	E $f(r)$	F $x(i+1)$	G $f(i+1)$	H $f(l)f(i+1)$
1	1	4	$= (2 - B2 + \ln(B2))$				
2	$= \text{IF}(H1 > 0, F2, B2)$	$= \text{IF}(H1 > 0, C2, F2)$					

General form: $\text{IF}(\text{logic}, [\text{if true}], [\text{if false}])$.

- Column B evaluation: says if value of f at $x(l)$ & at $x(i+1)$ are not oppositely signed update $x(l)$ by $x(i+1)$.
- Column C evaluation: says if value of f at $x(l)$ & at $x(i+1)$ are not oppositely signed keep $x(r)$ or else update.

The x_l, x_r notations are usual place holders for left & right end pts in bracketing methods and $x(i+1)$ is simply $(B1 + C1)/2$.

Once this is done rest is two drags away. Drag down D, E, F, G, H down to fill row 3. Then Shift+left click 8 times to select all of 3rd row & drag down 15–20 more iterations.

We stop according to given error margin. Suppose error margin is 10^{-4} . We will see our 14th error is 1.8310547×10^{-4} and 15th is 9.15527×10^{-5} . So, 15th $x(n+1)$ is our root.

Figure 1.8: Sign change across a discontinuity without a root

Drawback: Other than the rate of convergence the only other drawback is discontinuities and hence we started with continuous f to begin with, i.e., for discontinuous functions naturally we can have change in sign but no roots. In our diagram, bisection method will give us α as a root too. Once we get α , we have to check if α is a point of continuity to conclude anything.

1.3.2 Regula-Falsi method

We will start with a function f and find two initial guesses x_1 & x_2 such that

$$f(x_1)f(x_2) < 0$$

We will join $(x_1, f(x_1))$ to $(x_2, f(x_2))$ via straight line. Then we find where the straight line cuts the x -axis. We will come to the formulae later—first let's write down the scheme for Regula-Falsi. If

$$f(x_0)f(x_1) \begin{cases} > 0 & \text{use } x_0 \ \& \ x_2 \\ < 0 & \text{use } x_0 \ \& \ x_1 \end{cases}$$

Again draw the line joining $(x_0, f(x_0))$ & $(x_2, f(x_2))$ or $(x_1, f(x_1))$ & $(x_0, f(x_0))$.

Formulae for the algorithm: We did not assume f' exists.

$$\begin{aligned} \frac{y - y_1}{x - x_1} &= \frac{y_2 - y_1}{x_2 - x_1} \implies \frac{y - f(x_1)}{x - x_1} = \frac{f(x_2) - f(x_1)}{x_2 - x_1} \\ \text{Algorithm} &:= \frac{y - f(x_l)}{x - x_l} = \frac{f(x_r) - f(x_l)}{x_r - x_l} \stackrel{\text{def}}{=} \frac{y - f(x_r)}{x - x_r} = \frac{f(x_r) - f(x_l)}{x_r - x_l} \end{aligned}$$

where x_l & x_r are standard place holders for left & right end points in bracketing methods, and where x_l & x_r are either updated or maintained based on sign of $f(x_0^{(i)})f(x_l)$, & lastly where $x_0^{(i)} := i^{\text{th}}$ guess for root is achieved by solving the above equation for x using $y = 0$, i.e.,

$$\begin{aligned} -\frac{f(x_l)}{x - x_l} &= \frac{f(x_r) - f(x_l)}{x_r - x_l} \\ \implies -\frac{f(x_l)}{f(x_r) - f(x_l)} \cdot (x_r - x_l) &= x - x_l \\ x_0^{(m)} = x_l - \left(\frac{x_l - x_r}{f(x_l) - f(x_r)} \right) f(x_l) &\quad \text{OR} \quad x_0^{(m)} = x_r - \left(\frac{x_l - x_r}{f(x_l) - f(x_r)} \right) f(x_r) \end{aligned}$$

Remark 1.9. We will see Newton's method & Secant method that will have very similar form BUT there we will be starting with a single initial guess. Does that matter?

Yes! Regula-Falsi guarantees that we will converge to a root being one of the bracketed methods. That is not the case for Newton / Secant methods. However, we use Newton's & Secant's methods because they are much faster than Bisection & Regula-Falsi.

Both the above equations are the same expression once you multiply them out.

$$\boxed{x_0^{(m)} = \frac{x_r f(x_l) - x_l f(x_r)}{f(x_l) - f(x_r)}} \tag{1.9}$$

We write

$$|\mathcal{E}_{k+1}| = C|\mathcal{E}_k|^p \quad \text{where } \mathcal{E}_i := i^{\text{th}} \text{ error}$$

to denote a process has p -order rate of convergence. For Bisection, $C = \frac{1}{2}$ & $p = 1$ as seen already. Let's do it for Regula-Falsi.

When we get close to true root α , one of the end points gets fixed & the other end closes in on the true root. Let x_l be fixed after a few iterations and $x_r \rightarrow \alpha$. Then, we want to study how $\{x_n, x_{n+1}, \dots\}$ moves towards α . Let

$$\begin{aligned} x_l &\equiv a \\ x_k &= \alpha + \mathcal{E}_k \end{aligned}$$

Figure 1.9: One-sided endpoint freeze in Regula-Falsi

In our diagram, since right end point is moving we use the second form of Regula-Falsi involving x_r for obvious reasons because now x_{n+1} & x_n will both appear here in the equation:

$$\begin{aligned} x_{n+1} &= x_n - \frac{a - x_n}{f(a) - f(x_n)} f(x_n) \\ \implies \mathcal{E}_{n+1} &= \mathcal{E}_n - \frac{a - \alpha - \mathcal{E}_n}{f(a) - f(x_n)} f(x_n) \end{aligned}$$

Taylor Expansion around α gives us $f(x_n)$ that we can put in above equation:

$$\begin{aligned} f(x_n) &= f(\alpha + \mathcal{E}_n) = f(\alpha) + \mathcal{E}_n f'(\alpha) + \frac{1}{2} \mathcal{E}_n^2 f''(\alpha) + \dots \\ &= 0 + \mathcal{E}_n f'(\alpha) + \mathcal{O}(\mathcal{E}_n^2) \end{aligned}$$

$$\begin{aligned} \implies \lim_n \frac{\mathcal{E}_{n+1}}{\mathcal{E}_n} &= 1 - \lim_n \frac{(a - \alpha - \mathcal{E}_n)(\mathcal{E}_n f'(\alpha) + \mathcal{O}(\mathcal{E}_n^2))}{\mathcal{E}_n \{f(a) - \mathcal{E}_n f'(\alpha) - \mathcal{O}(\mathcal{E}_n^2)\}} \\ &= 1 - \lim_n \frac{(a - \alpha - \mathcal{E}_n)(f'(\alpha) + \mathcal{O}(\mathcal{E}_n^2)/\mathcal{E}_n)}{f(a) - \mathcal{E}_n f'(\alpha) - \mathcal{O}(\mathcal{E}_n^2)} \\ &= 1 - \frac{(a - \alpha)}{f(a)} f'(\alpha) := C. \end{aligned}$$

So, we get

$$\mathcal{E}_{n+1} = C |\mathcal{E}_n| \tag{1.10}$$

Geometric observations: If $f'(\alpha) \sim 0$, then, C is very close to 1. So, two consecutive errors do not differ much. The error will decay slowly. You can literally see this happen if you draw a function that has a double root or a slope very close to 0 at α as opposed to larger slopes as seen in 1.10.

Figure 1.10: Regula-Falsi for double root at α

Also note that, In order to get a constant C we did not have to use $|\mathcal{E}_n|^p$ —which we will need for Newton's method; in particular $p = 2$ will be needed which in turn will tell us that Newton's method is faster. Moreover, C here is dependent on a ; so we might have some f such that C is less than $\frac{1}{2}$ or greater than $\frac{1}{2}$. Hence, Regula-Falsi even though has same order as Bisection ($p = 1$), it can be faster or slower

than bisection. In practice, Regula-Falsi is faster since in practice we don't reach the zero, we go near it as much as needed for the sum & a process using secant lines will do that better.

Few important notes:

1. Rate of convergence analysis will always have similar structure as the one we just did. We will use the algorithm to get an equation in x_{n+1} & x_n , then put $\mathcal{E}_{n+1} + \alpha$ & $\mathcal{E}_n + \alpha$ to get an equation in $\mathcal{E}_{n+1}$ & $\mathcal{E}_n$. Finally, do Taylor expansion of f at some point (as needed by that particular sum) and get the expression $\mathcal{E}_{n+1} = C|\mathcal{E}_n|^p$ from which we can solve for the value of p .
2. Even though the processes don't need f'' to exist, in rate analysis we will assume f'' exists. We use f', f'' in Taylor expansion and this is fine; we need to get an idea of how good the algorithm is and for that we do not want it to be tainted by the bad-behaviour of the function itself. We simply need to pick a good test candidate on which all methods will be applied; such as the class of twice continuously differentiable functions.
3. Improvement of Regula-Falsi—*Illinois method*.

We had

$$\frac{x_r f(x_l) - x_l f(x_r)}{f(x_l) - f(x_r)}$$

for Regula-Falsi; now we are going to change weightage. If left end point is unchanged we will use $\frac{1}{2}f(x_l)$ for next iteration instead of $f(x_l)$.

Similarly, if x_r is fixed, use $\frac{1}{2}f(x_r)$. Then algorithm is

$$\frac{\frac{1}{2}x_r f(x_l) - x_l f(x_r)}{\frac{1}{2}f(x_l) - f(x_r)} \quad \text{OR} \quad \frac{x_r f(x_l) - \frac{1}{2}x_l f(x_r)}{f(x_l) - \frac{1}{2}f(x_r)}$$

Figure 1.11: Illinois method halving the function value at the stagnant point

As is natural from the diagram Illinois algorithm increases the odds of the next root ($n+1^{\text{th}}$ root) appearing on the other side than where x_n did. Of course, we may have x_n, x_{n+1} both appearing on the same side (easy to draw such graphs) but nevertheless, Illinois changes the odds.

Why? Because problem with Regula-Falsi was $(a - \alpha)$ appearing in C —the one side that was stuck forever was the issue. We only needed a way to get roots on both sides of α , then we close in on α from both sides. We will skip the calculation but this works! Order of convergence of Illinois is $1.442 > 1$!

Still not quadratic though but atleast better than linear.

4. This is just an observation. Note that $\frac{f(x_l) - f(x_r)}{x_l - x_r}$ is finite difference slope; this is the discrete version of f' we have seen in Mean Value Theorems. We use it a lot.

1.3.3 Newton's method

This is the fastest method but it requires f to be differentiable.

This is again an iterative method. Suppose we have x_n at hand & we want to find x_{n+1} . Draw a tangent at $f(x_n)$. The line meets x -axis at x_{n+1} . Let's find the formulae for the algorithm.

Figure 1.12: Newton-Raphson tangent line construction

Equation of the line: $\frac{y_1 - y_0}{x_1 - x_0} = m := \text{slope of the tangent line}$

$$\frac{f(x_n) - 0}{x_n - x_{n+1}} = f'(x_n) \implies \frac{f(x_n)}{f'(x_n)} = x_n - x_{n+1}$$

$$\boxed{x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}} \quad (1.11)$$

Repeat the process with x_{n+1} to get x_{n+2} as in the diagram and we see $\{x_n\}$ sequence is getting closer to the root.

Example 1.10. We will find $\sqrt{2}$. Let's start with $f(x) = x^2 - 2$.

$$f'(x) = 2x$$

Then,

$$x_{n+1} = x_n - \frac{x_n^2 - 2}{2x_n} = \frac{x_n^2 + 2}{2x_n}$$

Let's start with $x_0 = 1$.

$$x_1 = \frac{1 + 2}{2} = \frac{3}{2} = 1.5$$

$$x_2 = \frac{\frac{9}{4} + 2}{3} = \frac{17}{12} \approx 1.416$$

We put this in computer to do the calculation for various x candidates. Naturally, at any finite step we will be at a rational number because we are doing standard operations on rational numbers finitely many times & hence none will truly be $\sqrt{2}$; but the limit will be.

Example 1.11. We have seen all polynomial examples. Let's see $f(x) = a - \frac{1}{x}$ and use Newton.

$$f'(x) = \frac{1}{x^2}$$

Then,

$$\begin{aligned} x_{n+1} &= x_n - \frac{a - \frac{1}{x_n}}{\frac{1}{x_n^2}} \\ &= x_n - x_n^2 \left(a - \frac{1}{x_n} \right) \\ &= x_n - x_n^2 a + x_n \\ &= x_n(2 - ax_n) \end{aligned}$$

So,

$$x_{n+1} = x_n(2 - ax_n).$$

We start with some value of x_0 to get the true root here that is trivially $\frac{1}{a}$. Let $a = 2$.

Let $x_0 = 1$:

$$x_1 = 1(2 - 2 \cdot 1) = 0$$

$$x_2 = 0$$

$$\vdots \quad (\text{stuck at } 0)!!!!$$

This is a problem since we can never get to the true root $1/2$. What precisely went wrong will be seen later in rate of convergence discussion.

Let $x_0 = \frac{3}{4}$:

$$x_1 = \frac{3}{4} \left(2 - 2 \cdot \frac{3}{4} \right) = \frac{3}{4} \cdot \frac{1}{2} = \frac{3}{8} = 0.375$$

$$x_2 = \frac{3}{8} \left(2 - 2 \cdot \frac{3}{8} \right) = \frac{3}{8} \cdot \frac{5}{4} = \frac{15}{32} = 0.46875$$

$$x_3 = \frac{15}{32} \left(2 - 2 \cdot \frac{15}{32} \right) = \frac{15}{32} \cdot \frac{17}{16} = \frac{255}{512} \approx 0.498046875$$

$$x_4 = \frac{255}{512} \left(2 - 2 \cdot \frac{255}{512} \right) = \frac{255}{512} \cdot \frac{257}{256} \approx 0.49999237060546875$$

This is clearly converging to $\frac{1}{2}$.

We will now do the error analysis. Assume α is true root of f and the following three assumptions that will be used the proof for quadratic rate of convergence.

1. $f'(\alpha) \neq 0$ (α is not a multiple root).
2. $f''(x)$ must be continuous in a nbd around α .
3. Initial guess x_0 must be in the "basin of attraction".

We will again use Taylor expansion but notice that, this time it will not be around α , but, instead around x_k ; there is no particular reason other than the fact that once we see how the proof proceeds, we have to pick the point of evaluation smartly.

$$f(x) = f(x_k) + f'(x_k)(x - x_k) + \frac{f''(\xi_k)}{2}(x - x_k)^2 \quad (1.12)$$

where ξ_k lies between x_k & x .

Put $x = \alpha$ & in this case $x - x_k = \alpha - x_k$ is simply (true value) - (k^{th} approximation), hence, error. Let $\mathcal{E}_k = x_k - \alpha$ (we use $x_k - \alpha$ instead of $\alpha - x_k$ for ease of representation in next calculation). Put this in above equation and use $f(\alpha) = 0$ to get:

$$f(x_k) = f'(x_k)\mathcal{E}_k - \frac{f''(\xi_k)}{2}\mathcal{E}_k^2 \quad (1.13)$$

Again,

$$\begin{aligned}
\mathcal{E}_{k+1} &= x_{k+1} - \alpha \\
&= x_k - \frac{f(x_k)}{f'(x_k)} - \alpha \quad (\because \text{we are using Newton Method}) \\
&= \mathcal{E}_k - \frac{f(x_k)}{f'(x_k)} \\
&= \mathcal{E}_k - \frac{1}{f'(x_k)} \left\{ f'(x_k)\mathcal{E}_k - \frac{f''(\xi_k)}{2}\mathcal{E}_k^2 \right\} \quad (\text{we substituted equation 1.13})
\end{aligned}$$

So,

$$\mathcal{E}_{k+1} = \frac{f''(\xi_k)}{2f'(x_k)}\mathcal{E}_k^2 \quad (1.14)$$

$$\lim_{k \rightarrow \infty} \frac{|\mathcal{E}_{k+1}|}{|\mathcal{E}_k|^2} = \lim_{k \rightarrow \infty} \left| \frac{1}{2} \cdot \frac{f''(\xi_k)}{f'(x_k)} \right| = \frac{1}{2} \left| \frac{f''(\alpha)}{f'(\alpha)} \right| := C \quad (1.15)$$

$$(\because x_k \rightarrow \alpha \implies \xi_k \rightarrow \alpha).$$

With the two assumptions, equation 1.15 tells us that Newton's method has quadratic order of convergence.

If $f'(\alpha) \neq 0$, which is assumed, the above quantity is constant and this gives us $\mathcal{E}_{k+1} \sim \mathcal{E}_k^2$, the quadratic convergence rate we were talking about.

Also in equation 1.15, to take limit we must have the quantity

$$M_k = \left| \frac{1}{2} \frac{f''(\xi_k)}{f'(x_k)} \right|$$

to be finite & bdd. Hence, we need $f'(x_k) \neq 0$ as well, which is again one of the 3 assumptions.

Now we think about how to pick the initial point for starting our algorithm. If we are very far from true root α , then x_k might be such a point that acts as a stationary point for f , that is, slope of f at x_k is very close to 0 and $f'(x_k) \approx 0$! These are the scenarios where x_k is very far from α & f shows asymptotic behaviour. We must avoid these scenerios and pick close to α . However, this gives no proper quantification of how close we have to pick. For that we look at a very reasonable condition that errors must shrink, i.e.,

$$|\mathcal{E}_{k+1}| < |\mathcal{E}_k| \quad (1.16)$$

From equation 1.14 we have:

$$|\mathcal{E}_{k+1}| = \frac{1}{2} \left| \frac{f''(\xi_k)}{f'(x_k)} \right| |\mathcal{E}_k| |\mathcal{E}_k| \quad (1.17)$$

For equation 1.16 to be true we need $\frac{1}{2} \left| \frac{f''(\xi_k)}{f'(x_k)} \right| |\mathcal{E}_k| < 1$.

We call '**basin of attraction**' the interval around α from where x_0 can be picked & it is picked based on the condition:

$$\boxed{\frac{1}{2} \left| \frac{f''(\xi)}{f'(x)} \right| |x - \alpha| < 1} \quad (1.18)$$

Let us see now why $x_0 = 1$ failed for $f(x) = 2 - \frac{1}{x}$ in Example 1.11.

$$\begin{aligned} x_{n+1} &= x_n(2 - 2x_n) \\ \implies x_{n+1} - \frac{1}{2} &= -2 \left(x_n - \frac{1}{2} \right)^2 \\ \implies \mathcal{E}_{n+1} &= -2\mathcal{E}_n^2 \implies \frac{|\mathcal{E}_{n+1}|}{|\mathcal{E}_n|^2} = 2. \end{aligned}$$

We know,

$$\frac{1}{2} \left| \frac{f''(\xi_n)}{f'(x_n)} \right| = \frac{|\mathcal{E}_{n+1}|}{|\mathcal{E}_n|^2} = 2$$

Hence,

$$\frac{1}{2} \left| \frac{f''(\xi_n)}{f'(x_n)} \right| |x - \alpha| < 1 \iff 2|x - \alpha| < 1 \iff |x - \alpha| < \frac{1}{2}$$

Since for $a = 2$, true root $\alpha = \frac{1}{2}$, the condition gives x must be within $\pm \frac{1}{2}$ range of $\alpha = \frac{1}{2}$, i.e.,

$$x_0 \text{ must be in } \left(\frac{1}{2} - \frac{1}{2}, \frac{1}{2} + \frac{1}{2} \right) = (0, 1), \quad \text{i.e., } x_0 \in (0, 1)$$

So, $x_0 = 1$ is not allowed.

Drawback: Since the geometry of the process is very clear to us now, we can naturally guess that we need f to be nice enough for applying Newton's method. The algorithm itself needs f' to exist. Moreover, for getting basin of attraction margins we need further regularities such as $f \in \mathcal{C}^2$. Verify:

1. $f'(x) \neq 0$
2. $f \in \mathcal{C}^2$ or f' is Lipschitz (to avoid wild oscillations of the kind that $\sin(\frac{1}{x})$ shows)
3. α is an isolated root (avoid roots such as 0 for $x \sin \frac{1}{x}$ on $[0, 1]$).

Typical problems of Newton-Raphson's method:

1. When $f'(x_0) = 0$ for initial root guess. When we have 0 slope at one of the iterations, the slope never meets the x -axis & hence the process will break down, literally. We typically add a slight nudge β (tilts the 0 slope) to $f'(x_n)$ in this scenario to jumpstart the process. So, we get,

$$x_1 = x_0 + \frac{f(x_0)}{\beta}$$

2. When we are sent really far due to some slope. This happens when f'' is very high i.e., we have a sudden change in slope. This isn't a problem in particular, we will simply need a lot of iterations even though Newton Raphson is so fast.
3. We fall in a loop. We get same sequence of roots $\{x_0, x_1, \dots, x_k, x_0, x_1, \dots, x_k, x_0, x_1, \dots\}$. We have drawn a $\{x_0, x_1, x_0, x_1, \dots\}$ loop.

Figure 1.13: Failure modes of Newton-Raphson: zero slope, extreme slopes, and limit loops

All these are problems due to tangent line and they will appear in any slope based method. They were there in Regula Falsi and will be there in Secant method (we will see this next) too. Obviously, problem 2 of being sent really far can't happen in Regula-Falsi due to bracketing.

Remark 1.12. *Since for bisection method the only drawback is speed & discontinuity, we can for $f \in C^2$ use a mixed process. Use bisection to zoom in to the roots whatever they are (in general α is not known, hence, basin of attraction can't be calculated) and then after a few iterations (say 30–40 and we are at $\frac{1}{2^{30}}$ or $\frac{1}{2^{40}}$ error margin) we can use Newton.*

1.3.4 Secant method

Even though Newton's method is fastest, one of huge drawbacks is how f needs to be differentiable. This process is the next best option where if we do not have differentiability since we use Secant lines to approximate the tangent line.

We will start with two points x_1 & x_2 . Draw a line through $(x_1, f(x_1))$ & $(x_2, f(x_2))$. Get the point x_3 where line intersects x -axis. Up to this point this is same as Regula Falsi. Regula Falsi did several calculations at this point to decide if x_3 will replace x_1 or x_2 ; Secant will not check anything. Secant will always, as a rule, discard oldest value x_1 .

The pattern is always going to be $(x_1, x_2) \rightarrow (x_2, x_3) \rightarrow (x_3, x_4) \dots$

We will simply use "finite-difference slope" in Newton method.

$$x_{n+1} = x_n - f(x_n) \left(\frac{x_n - x_{n-1}}{f(x_n) - f(x_{n-1})} \right) \quad (1.19)$$

Figure 1.14: Secant method line intersection step

Remark 1.13. *We use this method a lot even if function is C^1 since just because $f'(x)$ exists doesn't mean it is easy to calculate for computer; in fact finite difference representations will be studied in details when we get to Numerical differentiation. Rate of convergence is 1.62 (Golden ratio) $\sim$ faster than linear but slower than Newton.*

Process of the proof is very similar to Newton's.

Assume $f \in C^2$ as always. If not, the following rate calculation will break down & it may still converge but at different rates depending on how rough f is. But, as established, for rate of convergence analysis we pick nice enough f so that we can purely judge the process that is not being affected by bad properties of the function.

$$\mathcal{E}_{n+1} = x_{n+1} - \alpha = x_n - f(x_n) \frac{x_n - x_{n-1}}{f(x_n) - f(x_{n-1})} - \alpha$$

$$\mathcal{E}_{n+1} = \mathcal{E}_n - f(x_n) \frac{x_n - x_{n-1}}{f(x_n) - f(x_{n-1})} \quad (1.20)$$

Note,

$$\begin{aligned} x_n &= \alpha + \mathcal{E}_n \\ x_{n-1} &= \alpha + \mathcal{E}_{n-1} \\ \implies x_n - x_{n-1} &= \mathcal{E}_n - \mathcal{E}_{n-1} \quad (\text{Substitute in numerator}) \end{aligned}$$

Then,

$$\begin{aligned} \mathcal{E}_{n+1} &= \mathcal{E}_n - f(x_n) \frac{\mathcal{E}_n - \mathcal{E}_{n-1}}{f(x_n) - f(x_{n-1})} = \frac{\mathcal{E}_n f(x_n) - \mathcal{E}_n f(x_{n-1}) - f(x_n) \mathcal{E}_n + f(x_n) \mathcal{E}_{n-1}}{f(x_n) - f(x_{n-1})} \\ \mathcal{E}_{n+1} &= \frac{\mathcal{E}_{n-1} f(x_n) - \mathcal{E}_n f(x_{n-1})}{f(x_n) - f(x_{n-1})} \end{aligned} \quad (1.21)$$

Here is the main distinction from Newton's method: We will do Taylor Series expansion of f around α . Earlier we had one moving point x_k & we expanded around x_k but now there are two moving points x_n & x_{n-1} and hence it is good for us to do the expansion around a fixed point α . We will in fact see cancellation in the numerator of 1.21 only if expansion is done around fixed pt α .

In fact, we will also not use Cauchy's form of Taylor expansion since if we do that we will have two mid points ξ_n and ξ_{n-1} and in that scenario we will have $f''(\xi_n)$ & $f''(\xi_{n-1})$ and this might hinder further cancellation in numerator.

Remark 1.14. *As you must have realized by now there isn't a fixed rule book about how to apply Taylor. It really depends on the problem itself. You will have 2 – 3 points around which expansion might seem useful, so, simply calculate and see what works.*

Taylor expansion around α :

$$f(x) = f(\alpha) + f'(\alpha)(x - \alpha) + \frac{f''(\alpha)}{2}(x - \alpha)^2 + \mathcal{O}(\mathcal{E}_k^3) \quad (1.22)$$

Let $x = x_n$ & then x_{n-1} in above expression; also use $f(\alpha) = 0$.

$$f(x_n) = f'(\alpha)\mathcal{E}_n + \frac{f''(\alpha)}{2}\mathcal{E}_n^2 + \mathcal{O}(\mathcal{E}_k^3) \quad (1.23)$$

$$f(x_{n-1}) = f'(\alpha)\mathcal{E}_{n-1} + \frac{f''(\alpha)}{2}\mathcal{E}_{n-1}^2 + \mathcal{O}(\mathcal{E}_k^3) \quad (1.24)$$

We put this in 1.21.

• Numerator:

$$\begin{aligned} \mathcal{E}_{n-1}f(x_n) - \mathcal{E}_n f(x_{n-1}) &= f'(\alpha)\mathcal{E}_n\mathcal{E}_{n-1} + \frac{f''(\alpha)}{2}\mathcal{E}_n^2\mathcal{E}_{n-1} - f'(\alpha)\mathcal{E}_{n-1}\mathcal{E}_n - \frac{f''(\alpha)}{2}\mathcal{E}_{n-1}^2\mathcal{E}_n + \mathcal{O}(\mathcal{E}_k^3) \\ &= \frac{\mathcal{E}_n\mathcal{E}_{n-1}}{2}f''(\alpha)\{\mathcal{E}_n - \mathcal{E}_{n-1}\} + \mathcal{O}(\mathcal{E}_k^3) \end{aligned}$$

• Denominator:

$$\begin{aligned} f(x_n) - f(x_{n-1}) &= f'(\alpha)\{\mathcal{E}_n - \mathcal{E}_{n-1}\} + \frac{f''(\alpha)}{2}\{\mathcal{E}_n^2 - \mathcal{E}_{n-1}^2\} + \mathcal{O}(\mathcal{E}_k^3) \\ &= (\mathcal{E}_n - \mathcal{E}_{n-1}) \left(f'(\alpha) + \frac{f''(\alpha)}{2}(\mathcal{E}_n + \mathcal{E}_{n-1}) \right) + \mathcal{O}(\mathcal{E}_k^3) \end{aligned}$$

Putting these in 1.21:

$$\begin{aligned} \mathcal{E}_{n+1} &= \frac{\mathcal{E}_n \mathcal{E}_{n-1}}{2} f''(\alpha)(\mathcal{E}_n - \mathcal{E}_{n-1}) \cdot \frac{1}{(\mathcal{E}_n - \mathcal{E}_{n-1}) (f'(\alpha) + f''(\alpha)(\mathcal{E}_n + \mathcal{E}_{n-1})/2)} \\ \Rightarrow \mathcal{E}_{n+1} &= \frac{f''(\alpha)}{\left(\frac{f'(\alpha) + f''(\alpha)(\mathcal{E}_n + \mathcal{E}_{n-1})}{2} \right) \cdot 2} \cdot \mathcal{E}_n \mathcal{E}_{n-1} \\ &= \frac{1}{\underbrace{2 \frac{f'(\alpha)}{f''(\alpha)} + (\mathcal{E}_n + \mathcal{E}_{n-1})}_{\equiv C}} \cdot \mathcal{E}_n \mathcal{E}_{n-1} \\ \lim_{n \rightarrow \infty} |C| &= \frac{1}{2} \cdot \left| \frac{f''(\alpha)}{f'(\alpha)} \right| \equiv C \quad (\because \mathcal{E}_n, \mathcal{E}_{n-1} \rightarrow 0) \end{aligned}$$

So,

$$\mathcal{E}_{n+1} = |C| \mathcal{E}_n \mathcal{E}_{n-1} \quad (1.25)$$

This is the most crucial observation that ends up giving the $1 + \frac{1}{p} = p$ equation in the exponent that we are about to see.

Next, to find rate of convergence we look for A & p such that

$$|\mathcal{E}_{n+1}| = A |\mathcal{E}_n|^p \quad \forall n \quad (*)$$

$$\begin{aligned} \Rightarrow |\mathcal{E}_n| &= A |\mathcal{E}_{n-1}|^p \quad \forall n \\ \Rightarrow |\mathcal{E}_{n-1}| &= \left(\frac{|\mathcal{E}_n|}{A} \right)^{\frac{1}{p}} \quad \forall n \end{aligned} \quad (**)$$

Put (*) & (**) in the error identity 1.25 for Secant method:

$$\begin{aligned} A |\mathcal{E}_n|^p &= |C| |\mathcal{E}_n| \left(\frac{|\mathcal{E}_n|}{A} \right)^{\frac{1}{p}} = |C| A^{-\frac{1}{p}} |\mathcal{E}_n|^{1+\frac{1}{p}} \\ \Rightarrow A^{1+\frac{1}{p}} |\mathcal{E}_n|^p &= |C| |\mathcal{E}_n|^{1+\frac{1}{p}} \end{aligned}$$

On comparing, $|C| = A^{1+\frac{1}{p}}$ & $p = 1 + \frac{1}{p} \Rightarrow p = \frac{1+\sqrt{5}}{2} \sim 1.61803398$.

We will put these in (*). C is known $\Rightarrow A = |C|^{(1+\frac{1}{p})^{-1}}$

$$|\mathcal{E}_{n+1}| = |C|^{1-\frac{1}{p}} |\mathcal{E}_n|^p \quad (1.26)$$

$$\text{where } |C| = \frac{1}{2} \cdot \left| \frac{f''(\alpha)}{f'(\alpha)} \right|, p = 1.61803398 \quad (1.27)$$

Similar to before the condition for a good initial guess is given via $|\mathcal{E}_{n+1}| < |\mathcal{E}_n|$ and this will give us our basin of attraction. Naturally for $|\mathcal{E}_{n+1}| < |\mathcal{E}_n|$ we need:

$$|C|^{1-\frac{1}{p}} |\mathcal{E}_n|^{p-1} < 1$$

$$\Rightarrow \left(\frac{1}{2} \left| \frac{f''(\alpha)}{f'(\alpha)} \right| \right)^{1-\frac{1}{p}} |x - \alpha|^{p-1} < 1$$

Basin of attraction for secant method consists of those x around α such that

$$|x - \alpha| < \left(\frac{1}{2} \left| \frac{f''(\alpha)}{f'(\alpha)} \right| \right)^{-\frac{1-1/p}{p-1}} \quad (1.28)$$

1.3.5 Fixed point iteration

This is usable only when we have $f(x) = 0$ and f has the following structure

$$f(x) = x - g(x)$$

We will use $x = g(x)$ as an update equation. Hence, it will be similar to jumping from one curve to another to either close in or diverge from the point where $x = g(x)$.

Assume initial value x_0 . Put it in $g(x_0)$. We continue doing this by treating $g(x_0)$ as input point. So, the algorithm is simply $g(g(\dots g(x_0) \dots))$.

Remark 1.15. *What if we have two ways to get $x - g(x)$ expression from $f(x)$? For example $2 - x + \ln(x) = f(x)$ can be represented as $x = 2 + \ln(x)$ or $x = e^{x-2}$. This example might be a bit misleading; they are inverse operations of each other, but we may have 3 representations too.*

Take $f(x) = x^3 - x - 1$:

$$x = (x + 1)^{\frac{1}{3}} := g_1(x)$$

$$x = \frac{1}{x^2 - 1} := g_2(x)$$

$$x = \frac{1}{x} + \frac{1}{x^2} := g_3(x)$$

They are all different curves of course, but they intersect $h(x) = x$ curve at the same point. It seems from figure 1.15 the rate of convergence is heavily dependent on the choice of g , and if that is the case this is a very erratic process.

Figure 1.15: Multiple fixed-point curves intersecting the line $y = x$

Let us more closely inspect the rate of convergence now. Let $\alpha = g(\alpha)$ (or let α be a root of f). As always assume $f \in \mathcal{C}^2 \Rightarrow g \in \mathcal{C}^2$.

Algorithm: $x_{n+1} = g(x_n)$.

Subtract and apply the Mean Value Theorem:

$$\begin{aligned}\mathcal{E}_{n+1} &:= \alpha - x_{n+1} = g(\alpha) - g(x_n) = g'(\xi)(\alpha - x_n) \\ \implies \mathcal{E}_{n+1} &= g'(\xi)\mathcal{E}_n\end{aligned}$$

$$\lim_{n \rightarrow \infty} \left| \frac{\mathcal{E}_{n+1}}{\mathcal{E}_n} \right| = |g'(\xi)| := C(g, \xi)$$

So, we have linear order, but C depends on g greatly! Having a faster result will come down to finding g such that g' is really small at $\xi \sim \alpha$ (flat around the true root α). In our above diagram 1.15 in terms of a faster α -finding function, we have the order

$$(x+1)^{\frac{1}{3}} \geq \frac{1}{x} + \frac{1}{x^2} \geq \frac{1}{x^2-1}$$

since $(x+1)^{\frac{1}{3}}$ has the least slope and $\frac{1}{x^2-1}$ has the most.

Convergence / Divergence:

$$\begin{aligned}\mathcal{E}_{n+1} &= C\mathcal{E}_n = C^2\mathcal{E}_{n-1} = \dots = C^n\mathcal{E}_1 \\ \lim_{n \rightarrow \infty} \mathcal{E}_{n+1} &= \lim_{n \rightarrow \infty} C^n\mathcal{E}_1 \rightarrow 0 \quad \text{if } |C| < 1.\end{aligned}$$

Fixed Point iteration converges if $f(x) = 0$ can be represented as $x = g(x)$ for g with the property $|g'(x)| < 1$ for all $x \in I \approx \text{nbhd of } \alpha$. Since in reality α will be unknown, we simply have to keep in mind that we initiate the method in an open set $\mathcal{U} \subseteq \mathbb{R}$ where $|g'| < 1$.

Figure 1.16: Monotonic versus oscillatory fixed-point iteration behavior

We also have an interpretation for $g' \in (0, 1)$ & $g' \in (-1, 0)$. When $g' \in (0, 1)$ we simply get to our point of intersection from one side, but when $g' \in (-1, 0)$ we get to the solution in a cyclic (spiral) manner. This is very natural too since the slope of $h(x) = x$ is fixed to be 1. If g has slope in $(0, 1)$ it is aligned with h and hence we will see monotonic staircase steps. If slope is in $(-1, 0)$ then it opposes what h looks like, and hence positive/negative error sign oscillations happen. What we have in figure 1.16 drew is what typically happens.

1.3.6 Higher dimensional equations

We already saw they have structure

$$f(x_1, \dots, x_n) = (f_1(\vec{x}), \dots, f_m(\vec{x})) = (0, \dots, 0) \in \mathbb{R}^m$$

We covered $n = m = 1$ case so far. We address higher dimensions now. We will not see $m < n$ or $m > n$ cases, only $m = n$ case. There too we will see two methods: Extension of Fixed Point Iteration and Extension of Newton Raphson.

1. Multivariable Fixed point iteration

Algorithm:

$$\begin{cases} x_1 = g_1(x_1, \dots, x_n) \\ \vdots \\ x_n = g_n(x_1, \dots, x_n) \end{cases} \sim \vec{x} = \vec{g}(\vec{x})$$

$$\begin{cases} x_1^{(k+1)} = g_1(x_1^{(k)}, \dots, x_n^{(k)}) \\ \vdots \\ x_n^{(k+1)} = g_n(x_1^{(k)}, \dots, x_n^{(k)}) \end{cases} \sim \vec{x}^{(k+1)} = \vec{g}(\vec{x}^{(k)})$$

$\vec{\mathcal{E}}^{(k)} = (\mathcal{E}_1^{(k)}, \dots, \mathcal{E}_n^{(k)})$ where $\mathcal{E}_j^{(k+1)} = |x_j^{(k+1)} - x_j^{(k)}|$.

We would check

$$\max_{j=1, \dots, n} \{\mathcal{E}_j^{(k)}\} := \|\vec{\mathcal{E}}^{(k)}\|_{\ell^\infty} < \text{given tolerance value.}$$

Establishing the relation between $\vec{\mathcal{E}}^{(k)}$ and $\vec{\mathcal{E}}^{(k+1)}$ for error analysis:

As always we will assume true root $\vec{\alpha} = (\alpha_1, \dots, \alpha_n)$ of $\vec{f}$ is known and define

$$\mathcal{E}_j^{(k)} := x_j^{(k)} - \alpha_j$$

Multivariable Taylor Expansion of $\vec{g}$ around $\vec{\alpha}$ is considered:

$$\begin{aligned} \vec{g}(\vec{x}^{(k)}) &= \vec{g}(\vec{\mathcal{E}}^{(k)} + \vec{\alpha}) = \vec{g}(\vec{\alpha}) + [J_{\vec{g}}(\vec{\alpha})]_{n \times n} \vec{\mathcal{E}}^{(k)} + \mathcal{O}(\|\vec{\mathcal{E}}^{(k)}\|^2) \\ \implies \vec{x}^{(k+1)} &= \vec{g}(\vec{x}^{(k)}) = \vec{\alpha} + J_{\vec{g}}(\vec{\alpha}) \vec{\mathcal{E}}^{(k)} + \mathcal{O}(\|\vec{\mathcal{E}}^{(k)}\|^2) \\ \implies \vec{x}^{(k+1)} - \vec{\alpha} &= J_{\vec{g}}(\vec{\alpha}) \vec{\mathcal{E}}^{(k)} + \mathcal{O}(\|\vec{\mathcal{E}}^{(k)}\|^2) \\ \implies \vec{\mathcal{E}}^{(k+1)} &= J_{\vec{g}}(\vec{\alpha}) \vec{\mathcal{E}}^{(k)} + \mathcal{O}(\|\vec{\mathcal{E}}^{(k)}\|^2) \end{aligned}$$

If we want to get rid of the $\mathcal{O}(\cdot)$ term we will have to use multivariate Mean Value Theorem and that will make things more complicated. For 1-variable ($n = 1$) we used MVT and hence no $\mathcal{O}(\cdot)$ term showed up; the remaining parts match.

Convergence would now boil down to some condition on the matrix $J_{\vec{g}}(\vec{\alpha})$:

$$\vec{\mathcal{E}}^{(k+1)} \approx J_{\vec{g}}(\vec{\alpha}) \vec{\mathcal{E}}^{(k)} = J_{\vec{g}}(\vec{\alpha}) J_{\vec{g}}(\vec{\alpha}) \vec{\mathcal{E}}^{(k-1)} = \dots = J_{\vec{g}}(\vec{\alpha})^k \vec{\mathcal{E}}^{(1)}$$

$$\lim_{k \rightarrow \infty} \|\vec{\mathcal{E}}^{(k+1)}\| = \lim_{k \rightarrow \infty} \|J_{\vec{g}}(\vec{\alpha})^k \vec{\mathcal{E}}^{(1)}\| = 0$$

only when spectral radius of matrix $J_{\vec{g}}(\vec{\alpha})$ are strictly less than 1 in magnitude.

Remark 1.16. Starting with $\vec{f}(\vec{x}) = 0$, we turn it into an iterative scheme by:

$$\vec{g}(\vec{x}) = \vec{x} + \beta \vec{f}(\vec{x}) = \vec{x}$$

β is a scalar that helps with the spectral radius condition of $J_{\vec{g}}$.

2. Multivariable Newton's method

$$\begin{cases} f_1(x_1, \dots, x_n) = 0 \\ \vdots \\ f_n(x_1, \dots, x_n) = 0 \end{cases}$$

Recall single variable case:

$$x^{(k+1)} = x^{(k)} - \frac{f(x^{(k)})}{f'(x^{(k)})} \quad (1.29)$$

$$\implies f(x) = f(x^{(k)}) + f'(x^{(k)})(x - x^{(k)}) \quad (1.30)$$

In the second form we can clearly see the Taylor Expansion at $x = x^{(k+1)}$. We do the same linearization to each equation (but now with n -many derivatives involved):

$$f_j(x_1, \dots, x_n) := f_j(\vec{x}) = f_j(\vec{x}^{(k)}) + \sum_{i=1}^n \frac{\partial f_j}{\partial x_i} \bigg|_{\vec{x}^{(k)}} (x_i - x_i^{(k)}) \quad \text{for } j = 1, \dots, n$$

where $\vec{x}$ is a variable and $\vec{x}^{(k)}$ is the k^{th} guess.

Again compare with the single variable case. Apriori, our new observation makes $f = 0$. Here too we will do that, but it is not so easy now due to $[J_{\vec{f}}(\vec{x}^{(k)})]^{-1}$ that we have in this method:

$$\begin{aligned} \begin{bmatrix} f_1(\vec{x}) \\ \vdots \\ f_n(\vec{x}) \end{bmatrix} &= \begin{bmatrix} f_1(\vec{x}^{(k)}) \\ \vdots \\ f_n(\vec{x}^{(k)}) \end{bmatrix} + \begin{bmatrix} \partial_{x_1} f_1 & \dots & \partial_{x_n} f_1 \\ \vdots & \ddots & \vdots \\ \partial_{x_1} f_n & \dots & \partial_{x_n} f_n \end{bmatrix} \begin{bmatrix} x_1 - x_1^{(k)} \\ \vdots \\ x_n - x_n^{(k)} \end{bmatrix} \\ \implies \vec{f}(\vec{x}) &= \vec{f}(\vec{x}^{(k)}) + J_{\vec{f}}(\vec{x}^{(k)})(\vec{x} - \vec{x}^{(k)}) = \vec{f}(\vec{x}^{(k)}) + J_{\vec{f}}(\vec{x}^{(k)})\Delta\vec{x}^{(k)} \end{aligned}$$

Apriori, next guess $\vec{x}^{(k+1)}$ is what makes $\vec{f} = 0$.

We will simply solve by putting 0 on LHS, but this involves inverting the Jacobian. We get

$$J_{\vec{f}}(\vec{x}^{(k)})\Delta\vec{x}^{(k)} = -\vec{f}(\vec{x}^{(k)})$$

This is the usual linear system where we solve for $\Delta\vec{x}^{(k)}$, and once we get it we have our $\vec{x} = \Delta\vec{x}^{(k)} + \vec{x}^{(k)}$ that will be the new guess for the root.

The main **algorithm** of Multivariate Newton's method is:

$$\boxed{\vec{x}^{(k+1)} = \vec{x}^{(k)} - [J_{\vec{f}}(\vec{x}^{(k)})]^{-1} \vec{f}(\vec{x}^{(k)})} \quad (1.31)$$

A slight modification: When J^{-1} doesn't exist, i.e., $\det(J_{\vec{f}}(\vec{x}^{(k)})) = 0$, in that case we opt for a trick that we have seen during regression. Consider the same algorithm but with $(J^T J)^{-1} J^T$ instead of J^{-1} !

In fact, we do a further modification of adding a slight perturbation to this inversion system:

$$\boxed{\vec{x}^{(k+1)} = \vec{x}^{(k)} - (J^T J + \beta I_n)^{-1} J^T \vec{f}(\vec{x}^{(k)})} \quad (1.32)$$

As already stated the idea is similar to regression, hence in a more formal proof “least-squares” does show up. This is called the **Levenberg–Marquardt algorithm** of damped least squares. This is a very important idea and falls under non-linear regression. We will see a bit of non-linear fitting soon but we simply see ‘polynomial regression’, but suppose we want to fit ae^{bx} or $a \ln(bx) + a^2e^{by}$, etc., to a given set of data—then LM damped least squares is used.

DATA FITTING TECHNIQUES

2

2.1 Section 1

2.1.1 Subsection 1

NUMERICAL CALCULUS TECHNIQUES

3

3.1 Section 1

3.1.1 Subsection 1

NUMERICAL DIFFERENTIAL EQUATIONS

4